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Project Title:	Metastable Solution Structure and Optimization Strategies in Protein Crystal Growth		
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Performance Goal Text:	tain a leading biotechnology research program that will assure continued scientific and technical leadership.		
	First we will fully develop the crosslinking experiments using a variety of crosslinking reagents that are known to react rapidly under the acidic conditions of tetragonal lysozyme crystal growth. The crosslinking reaction will be used to study aggregates in solutions that lead to crystal growth of lysozyme and insulin. We will explore the temperature and salt dependence of the aggregate distributions. From the temperature dependence we will determine the thermodynamic parameters which will be used to understand the intermolecular forces which predominate under both crystallizing and precipitating conditions. We will apply simplex optimization techniques to protein crystallization by optimizing the conditions for crystal growth of the well-characterized systems of insulin and lysozyme. This approach will be applied to the crystallization of human serum transferrin, a more challenging crystallization problem. If simplex optimization leads to diffraction quality crystals of transferrin we plan to solve it’s monoferric structure using molecular replacement techniques. There are three primary objectives of our work:		

Task Description:	1) Fully characterize metastable solutions of proteins under solution conditions which lead to precipitate and crystallization. The presence and distribution of aggregates in supersaturated solutions will be investigated by chemical cross-linking followed by SDS-PAGE, size-exclusion chromatography and capillary electrophoresis. The role of precipitating agents in aggregate formation will be studied using different inorganic salts and polyethylene glycol. The time dependence of prenucleation clusters will be analyzed using turbidity analysis. 2) Determine if aggregation rates (aggregates formed as a function of time) are lowered in microgravity by carrying out chemical cross-linking experiments on Space Shuttle flights in the Fluid Processing Apparatus. If a decrease in the aggregation rate is found, then fewer, and possibly smaller, aggregates would be present during the growth stage. With fewer aggregates present we propose that smaller, more ordered, aggregates would be the favored growth units. It is possible that this, and a reduced transport of impurities or nonspecific aggregates to the surface, could be the explanation for the higher quality protein crystals seen in microgravity. 3) Develop simplex optimization techniques for protein crystal growth to be used by investigators doing microgravity crystallization experiments on the International Space Station. This approach requires fewer experiments than both the single-factor-at-a-time approach or factorial designs and will conserve material on Space Station. In addition, it will allow optimization of conditions for microgravity which are probably different than the optimum conditions for crystal growth on Earth. With the growing need for new medications to fight disease structure-based drug design is becoming more important. The first step in the process to design a new drug involves crystallization of an important biological macromolecule. Our work looks at the crystallization process in an attempt to further understand it and design new methods for crystal growth. This will have implications not only in drug design but also in the understanding of many biological mechanisms. The long-term objectives of the proposed research are to further the understanding of the fundamental processes of protein nucleation and crystal growth both on the ground and in microgravity. With the recent results from NASA's microgravity crystallization experiments we now plan to answer the following questions using our experimental techniques: 1) Does gravity affect the formation of aggregates? 2) Does gravity affect the size distribution and/or average size of the aggregates? 3) Do the aggregates play a different role in the crystal growth process in microgravity than on Earth? 4) Does gravity affect the transport of the aggregates to the crystal surface? Answers to these important questions will allow crystallographers to design crystallization methods that control aggregate formation thereby increasing the quality of crystals obtained on the ground and in microgravity. This will lead to better crystal structures which has implications in the fields of drug design, protein engineering, and many areas of biomedical research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work investigates the nucleation and growth of protein crystals, which are used to determine protein structures, and eventually leads to the development of new drugs.
Task Progress:	Three analytical methods, SDS-PAGE, SDS-CE and SEC, were used to separate protein aggregates formed by chemical cross-linking with glutaraldehyde. The first method, SDS-PAGE, is fast and provides direct evidence of reaction products. However, it requires densitometry for quantitation and is not very sensitive. SDS-CE is very sensitive and provides excellent resolution of the protein aggregates. It still requires SDS to impart a negative charge to produce the electroosmotic flow through a sieving gel in the run buffer. Separation is based primarily on size because a sieving gel is added to the capillary. The only method that does not require denaturation is SEC which gives a more accurate picture of the aggregate distribution in solution. SEC without cross-linking results in a single peak even at high salt concentrations. If higher order aggregates are present once they are separated from the monomer they dissociate during the separation to form more monomer to meet the equilibrium criteria. However, glutaraldehyde cross-linking adds a covalent bond to protein molecules whose e-amino nitrogen atoms are 7.5 Å apart which implies that their a-carbon atoms are no more than 20 Å apart. Since only 2 cross-links have been observed in crystallographic data, it is thought that the cross-linking leads to flexible chains which would not be readily observable in difference maps. Using the same cross-linking reaction mixture we obtained 52.53% monomer by SEC and 59.77% monomer by SDS-CE. This represents a 7.24% increase in the monomer when subjected to denaturation by SDS, ß-mercaptoethanol and boiling. This indicates that some of the higher order aggregates are broken down to monomer during denaturation. Cross-links, just like disulfide bonds, restrain the natural unfolding and folding of the protein. However, it has been shown that even proteins with unreduced disulfide bonds can still be converted to random coils by denaturant . Since monomers are being produced during denaturation, the cross-linking reagent appears to be cleaved by reduction. We also lowered the ratio of glutaraldehyde:protein from 1000:1 in our previous work to 500:1 in this work. Lower concentrations of glutaraldehyde have been shown to result in less stable cross-links. Using SEC we see evidence of the first three species in supersaturated CEWL solutions which is in agreement with this model. However, octamers are noticeably absent. Their absence may be due to the fact that all of the free lysine residues are saturated at the tetramer stage. Evidence for lack of free lysines is further supported by the low monomer molecular weight after cross-linking as determined by SEC which suggests a good deal of intramolecular cross-linking. All three separation methods show direct evidence of dimers in the absence of. This dimer is formed during the cross-linking process and is not the naturally occurring dimer of CEWL since non-cross-linked CEWL was monomeric. One explanation for the formation of such dimers is the polymerization of glutaraldehyde. At high glutaraldehyde concentrations the formation of nonspecific inter-aggregate cross-links can occur which could lead to dimer formation even at low ionic strength. An increase in the chain length of a glutaraldehyde cross-link could potentially join proteins quite far apart from each other. NMR studies have shown that commercial glutaraldehyde solutions contain the dialdehyde in equilibrium with its cyclic monohydrate and open chain monohydrate and dialdehyde. When the solution is near neutral or alkaline, the dialdehyde undergoes an intermolecular aldol condensation which is followed by dehydration to give a-ß-unsaturated aldehyde polymers, OHC-CH2-[CH2]2-[CH=C(CHO)-[CH2]2]n-CH=C(CHO)[CH2]2CHO. Commercial glutaraldehyde solutions of 25%, at pH 3, have been found to contain 79% water, 3% glutaraldehyde and 18% derivatives of higher molecular weight. The higher molecular weight derivatives of glutaraldehyde can react with proteins either in polymer form or by depolymerization. The polymerization of glutaraldehyde must be considered when interpreting the percent aggregates at higher ionic strength as well as the dimer formation at low ionic strength where lysozyme is known to be monomeric. It

	is possible that high ionic strength solutions increase the polymerization of glutaraldehyde allowing for higher aggregate concentrations. Because of the tendency to polymerize, glutaraldehyde cannot be used to determine intermolecular bond distances like more specific protein cross-linking agents. Also, to prevent polymers of glutaraldehyde from participating in the cross-linking it is important to keep the glutaraldehyde concentration relatively low. The age of the glutaraldehyde solution has been found to affect the extent of polymerization. For this reason, we used fresh glutaraldehyde solutions and stored the glutaraldehyde under nitrogen. SDS-PAGE was used to separate cross-linked product with different amounts of time between addition of salt and the start of cross-linking. One would expect to see a gradual increase in particle size with time. However, there was no difference seen in the percentage and types of aggregates. Given the results of SEC it is now clear that SDS-PAGE does not give an accurate picture of the aggregate distribution. Larger aggregates are broken up into smaller ones during the denaturation process. Therefore, even though this was the final year of the grant future work will focus on following the time dependence of aggregation using native SEC for separation.
Bibliography Type:	Description: (Last Updated: 02/10/2006)
Articles in Peer-reviewed Journals	Hall, C.L.; Clemens, J.R.; Brown, A. M. and Wilson, L.J. "Supersaturated Lysozyme Solution Structure Studied by Chemical Cross-linking" Acta Crystallographica Section D Vol. 61, 813–818 , Feb-2005
Presentation	Clemens, J.;Brown, A.;Hall, C.;Lanham, K.;Myers, G.;Norwitz, S.;Penn, W.; Harvey, C.;Wilson, L. "Report on Analysis of multi-planar mixing in a microgravity environment" 11th Annual EPSCoR Conference, Advances in Research Through EPSCoR, Lexington, KY May-2005
Presentation	Clemens, J., Brown, A., Hall, C., Lanham, K., Myers, G., Norwitz, S., Penn, W., Harvey, C., Wilson, L. "Multi-planar mixing in a microgravity environment" UP Showcase, Eastern Kentucky University, Richmond, KY 40475 Apr-2004